

Software Requirements Specification (SRS)

TheMoodShelf

Intelligent Mood-Based Book Recommendation System

Version: 1.1

Date: March 31, 2026

1. Introduction

1.1 Purpose

TheMoodShelf is a web application that detects user emotions from text input and recommends relevant books using ML models. It tracks mood history, allows journaling, provides analytics insights, and supports cloud hosting.

1.2 Scope

- Emotion classification using DistilRoBERTa
- Semantic book matching with Sentence Transformers
- User authentication and session management
- Mood history, journaling, and analytics
- Multi-book recommendation results

2. Functional Requirements

2.1 Core Features

ID	Requirement	Priority
F-001	User registration/login with hashed passwords	High
F-002	Free-text mood input with emotion classification (7 emotions)	High
F-003	Display top relevant books with similarity scores	High
F-004	Save books to personal library with mood context	High
F-005	Journal entries linked to mood entries	Medium
F-006	Analytics dashboard: mood trends over time, favorite genres, reading stats	Medium
F-007	Regenerate recommendations with variety randomization	Medium

2.2 Mood Detection Logic

- Aggregate emotions into 8 moods: happy, sad, stressed, calm, anxious, lonely, tired, motivated
- Keyword overrides for confidence < 0.65
- Default to calm if confidence < 0.25

2.3 Book Matching

- Filter by mood-relevant genres first
- Rank by cosine similarity between user text and book descriptions
- Return top results with metadata (title, author, image, rating, review count)

3. Non-Functional Requirements

Category	Requirement	Target/Status
Performance	Page load time	< 2s
	Mood analysis	< 3s
	Concurrent users	50+
Security	Password hashing (Werkzeug)	Required
	Parameterized SQL queries	Required
	HTTPS for production	Required
Reliability	99% uptime for hosted version	Target
	Graceful fallback if ML model fails	Required
Scalability	Cloud-deployable (Heroku/AWS/GCP)	Required
Usability	Mobile-responsive design	Required

4. Interface Requirements

4.1 User Interfaces

Page	Purpose
Login/Signup	Authentication
Dashboard	Mood input, quick stats
Results	Top 5 book recommendations with emotion breakdown
Saved Books	Personal library with filters
Journal	Write entries linked to current mood
Analytics	Mood trends (line charts), genre distribution, reading patterns

4.2 System Interfaces

- Database: MySQL (production), SQLite (development)
- ML Models: Hugging Face Transformers (DistilRoBERTa, all-MiniLM-L6-v2)
- Hosting: Compatible with Heroku, AWS Elastic Beanstalk, or GCP App Engine

5. Design Constraints

Constraint	Description
Tech Stack	Python 3.8+, Flask, MySQL, HTML/CSS/JS
ML Models	Fixed pre-trained models (384-dim embeddings)
Dataset	Static CSV (Book_Dataset_1.csv), latin1 encoding
Hosting	Must support Python WSGI, 1GB+ RAM for models
Storage	Cloud database required for hosted deployment

6. Data Model

Database tables:

- **users** (id, username, email, password_hash)
- **mood_entries** (id, user_id, mood, input_text, top_emotion, timestamp)
- **saved_books** (id, user_id, title, author, mood, saved_at)
- **journal_entries** (id, user_id, mood_entry_id, content, timestamp)

7. Future Enhancements (Post-Coursework)

- Real-time book API integration (Google Books/OpenLibrary)
- Social features (share recommendations)
- Mobile app version
- Advanced analytics (predictive mood trends)